EVE Agent Delegation

Real EO retrieval through LangGraph, A2A and MCP

A working prototype around the official EVE MCP tools. A supervisor plans an EO request, delegates retrieval across processes, and returns structured results with sources and trace.

27
automated tests passed
Protocol, validation and recovery

21.69s
LLM + live-data demo
Request-to-terminal time

4/4
LLM case expectations
Real model; fixture retrieval

ARCHITECTURE | actual process and protocol boundaries

Supervisor process
LangGraph + FastAPI

Plan and route
Delegate and recover
Assemble sourced results

SQLite checkpoints

A2A →

Retrieval process
Official A2A Python SDK

Discover Agent Card
Accept A2A task
Run MCP client over stdio

Return DataPart artifact

MCP →

External data
Official EVE MCP tools

GDACS
Disaster event reports

EO Dashboard
STAC catalogue resources

Results and source links return through A2A; both services run independently.

LIVE EXAMPLE

Request: Find a recent flood, identify EO resources, and explain their use for monitoring.
Retrieved: GDACS India and China flood reports plus an EO Dashboard alos2_floods collection candidate.

VERIFICATION | separate evidence classes

Rules planner: 21/21 offline case expectations; 2/3 live cases returned data. The third reported no matching catalogue resources.
Real LLM: DeepSeek V4.1 Flash selected tools and filters. Four fixture cases passed, including Chinese filters and rejection of unrelated requests.
Robustness: Bounded retries, failure/no-data handling and supervisor crash recovery were exercised. Locked dependencies and raw run records support reproduction.

Scope and interpretation
Catalogue candidates are not verified imagery for a disaster. LLM planning does not generate retrieved facts; monitoring guidance is deterministic. Small samples establish integration only. This is an independent demo, not an ESA release.

SOURCE + SETUP | LIVE RUN EVIDENCE | FULL EVALUATION

github.com/YunongDai2005/eve-agent-delegation-mvp